

SSILP Math

Beginner

INEQUALITIES
COORDINATE PLANE
GRAPHS OF LINES AND CIRCLES



Learning objectives

1. Understand and solve inequalities:

1. Represent inequalities on the number line.
2. Solve linear inequalities and absolute value inequalities.

2. Graph regions on the coordinate plane:

1. Sketch regions and apply the distance formula.
2. Graph circles and determine their center and radius.

3. Analyze graphs of lines:

1. Determine x- and y-intercepts.
2. Understand the concept of slope and measure the steepness of a line.
3. Find the equation of a line through given points and use slopes and y-intercepts to graph lines.

Session Outline

1. Inequalities

And other injustices

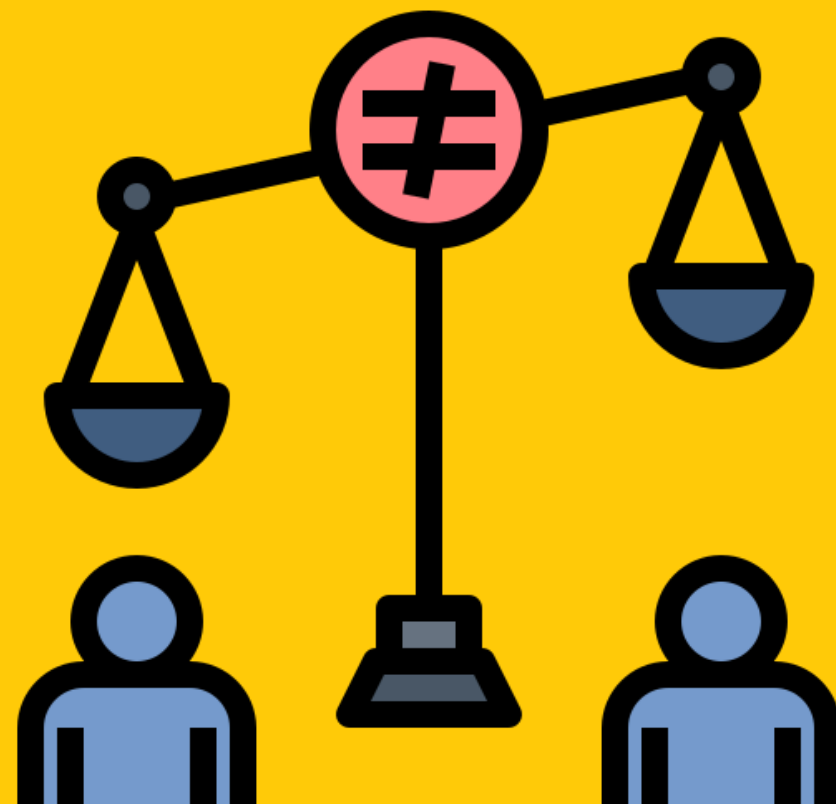


2. Coordinate plane, Graphs of lines and circles

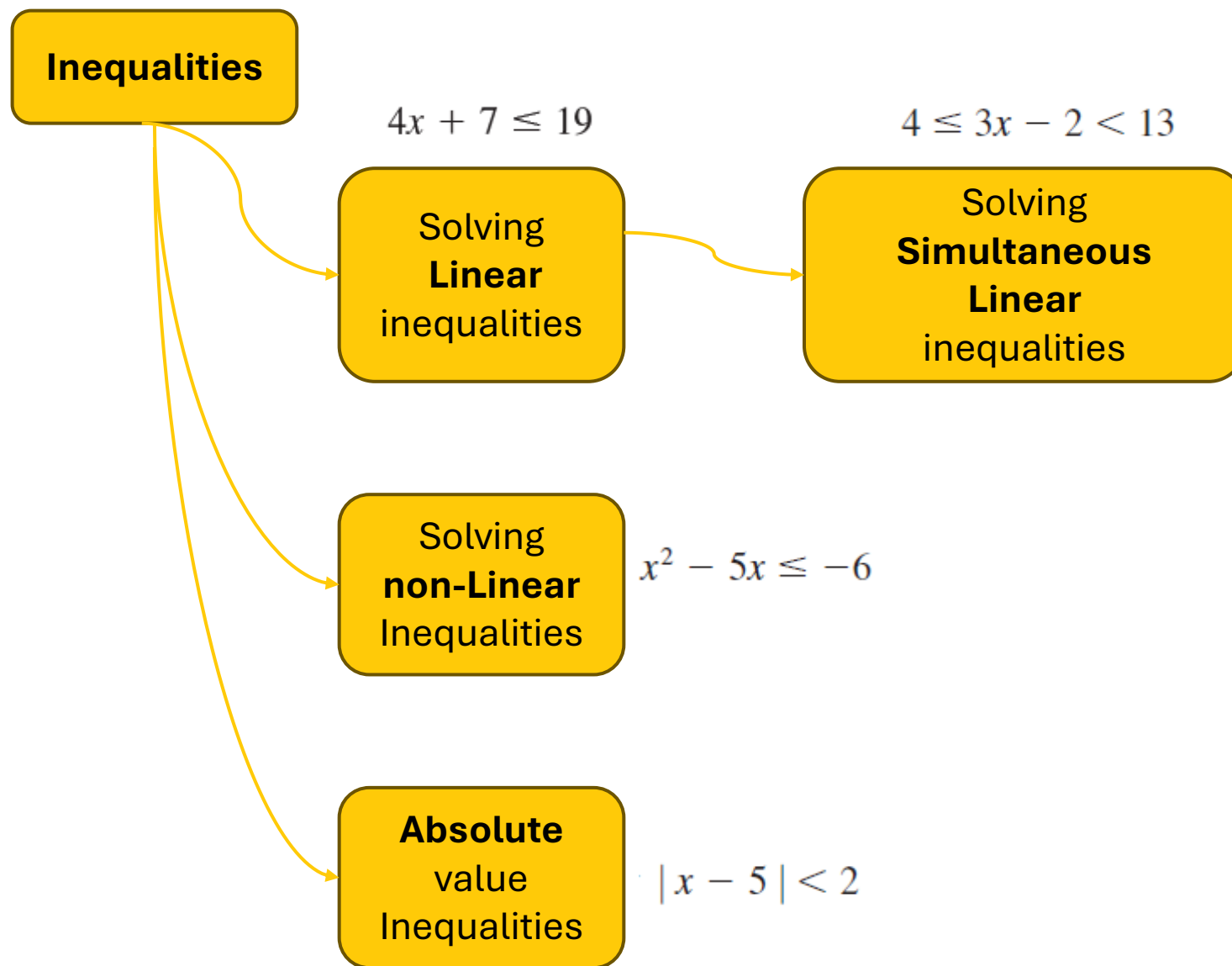


1. Inequalities

And other injustices



Inequalities concept map:



equations

$$5x - 2 = 8$$

$$5x = 10$$

$$x = 2$$

x has only **one value**

inequalities

$$x > 2$$

x is greater than 2

$$x < 2$$

x is less than 2

x has **many values**

Mnemonic.

The $<$ symbol looks like a slanted L which stands for "Less than".

Inequality

Here's an example

$$x > 3$$

x has many values

x can be **3.000001**

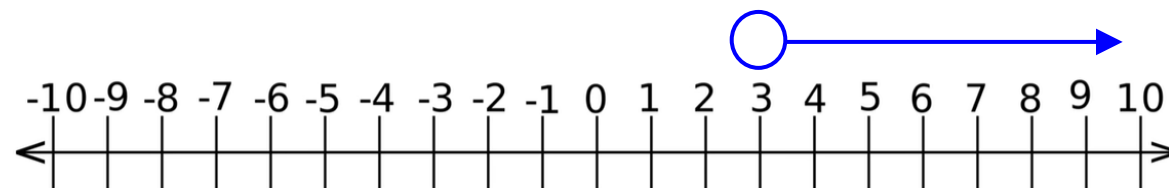
or **4**

or **10**

or **1000,000, 000**

Any number bigger than 3 is a **solution**

How we represent inequalities
on the number line:



the open circle means 3 is
excluded from the solution set

0, -3, 2.99999, 3 are not solutions

Solving inequalities

$$x + 2 < 7$$

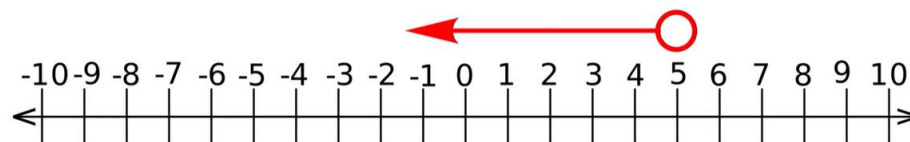
inequalities are solved
just like **equations**

$$x + 2 - 2 < 7 - 2$$

subtract 2
from both sides

$$x < 5$$

any value of x that is **less than 5**
is a solution to the inequality



Solving inequalities: Note this point!

Inequalities work similar to equations, but there is one **BIG DIFFERENCE**

if multiplying or dividing
by a negative number
we must reverse the sign
of the inequality

$$-x > 6$$

$$-1 \cdot -x > 6 \cdot -1$$

multiply or divide
both sides by -1
to get positive x

$$x < -6$$

Consider this

$$1 < 2 \quad \checkmark$$

If we multiply both
sides with -1, we get

$$-1 < -2 \quad \times$$

Correct way is to
reverse the sign

$$-1 > -2 \quad \checkmark$$

Greater than equal to/ Less than equal to

$$x \geq 5$$

x is **greater than or equal to** five

$$x \leq 5$$

x is **less than or equal to** five

Include the value of shown,
rather than excluding it.

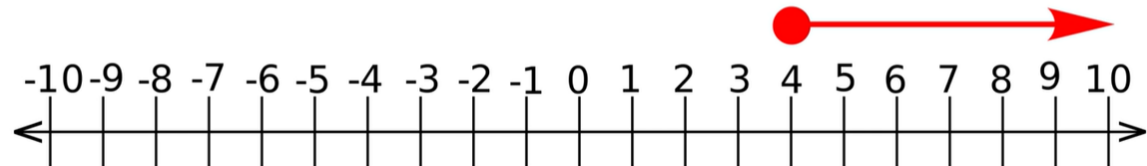
$$3x + 4 \geq 16$$

$$3x + 4 - 4 \geq 16 - 4$$

$$\frac{3x}{3} \geq \frac{12}{3}$$

$$x \geq 4$$

the closed circle means 4 is
included in the solution set



Inequality rules

RULES FOR INEQUALITIES

Rule

1. $A \leq B \Leftrightarrow A + C \leq B + C$
2. $A \leq B \Leftrightarrow A - C \leq B - C$
3. If $C > 0$, then $A \leq B \Leftrightarrow CA \leq CB$
4. If $C < 0$, then $A \leq B \Leftrightarrow CA \geq CB$
5. If $A > 0$ and $B > 0$,
then $A \leq B \Leftrightarrow \frac{1}{A} \geq \frac{1}{B}$
6. If $A \leq B$ and $C \leq D$,
then $A + C \leq B + D$
7. If $A \leq B$ and $B \leq C$, then $A \leq C$

Description

Adding the same quantity to each side of an inequality gives an equivalent inequality.

Subtracting the same quantity from each side of an inequality gives an equivalent inequality.

Multiplying each side of an inequality by the same *positive* quantity gives an equivalent inequality.

Multiplying each side of an inequality by the same *negative* quantity *reverses the direction* of the inequality.

Taking reciprocals of each side of an inequality involving *positive* quantities *reverses the direction* of the inequality.

Inequalities can be added.

Inequality is transitive.

Solving Linear inequalities

An inequality is **linear** if each term is constant or a multiple of the variable

Solve the inequality $3x < 9x + 4$, and sketch the solution set.

SOLUTION

$$3x < 9x + 4$$

Given inequality

$$3x - 9x < 9x + 4 - 9x$$

Subtract $9x$

$$-6x < 4$$

Simplify

$$\left(-\frac{1}{6}\right)(-6x) > \left(-\frac{1}{6}\right)(4)$$

Multiply by $-\frac{1}{6}$ and reverse inequality

$$x > -\frac{2}{3}$$

Simplify

The solution set consists of all numbers greater than $-\frac{2}{3}$. In other words the solution of the inequality is the interval $\left(-\frac{2}{3}, \infty\right)$. It is graphed in Figure 1.

FIGURE 1

EXAMPLE

Simultaneous, linear inequalities

Also known as
“Double” inequalities

EXAMPLE 2 ■ Solving a Pair of Simultaneous Inequalities

Solve the inequalities $4 \leq 3x - 2 < 13$.

SOLUTION The solution set consists of all values of x that satisfy both of the inequalities $4 \leq 3x - 2$ and $3x - 2 < 13$. Using Rules 1 and 3, we see that the following inequalities are equivalent:

$$4 \leq 3x - 2 < 13 \quad \text{Given inequality}$$

$$6 \leq 3x < 15 \quad \text{Add 2}$$

$$2 \leq x < 5 \quad \text{Divide by 3}$$

Therefore the solution set is $[2, 5)$, as shown in Figure 2.

FIGURE 2

EXAMPLE

Determine which elements satisfy the inequality

Let $S = \{-5, -1, 0, \frac{2}{3}, \frac{5}{6}, 1, \sqrt{5}, 3, 5\}$.

Determine which elements of S satisfy the inequality

(a) $1 - 2x \geq 5x$

(b) $-2 \leq 3 - x < 2$

ACTIVITY 1.1

1.1 Solving Inequalities

Solving linear inequalities

(a) $-4x \geq 10$

(b) $3x + 11 < 5$

(c) $5 - 3x \leq -16$

(d) $\frac{2}{5}x + 1 < \frac{1}{5} - 2x$

ACTIVITY 1.2

1.2a,b: Solving linear Inequalities

1.2c,d: Solving linear Inequalities

Absolute value inequality: Properties

PROPERTIES OF ABSOLUTE VALUE INEQUALITIES

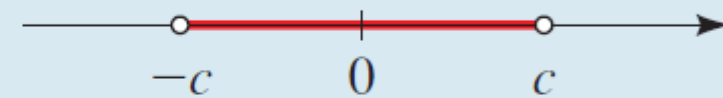
Inequality

Equivalent form

Graph

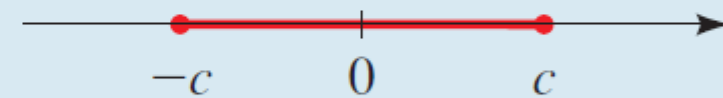
1. $|x| < c$

$-c < x < c$



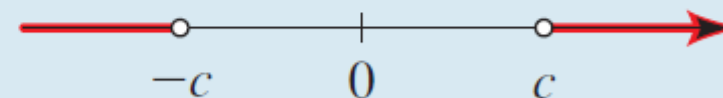
2. $|x| \leq c$

$-c \leq x \leq c$



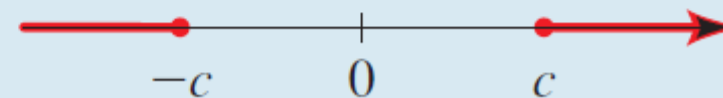
3. $|x| > c$

$x < -c$ or $c < x$



4. $|x| \geq c$

$x \leq -c$ or $c \leq x$



Solving an absolute value inequality

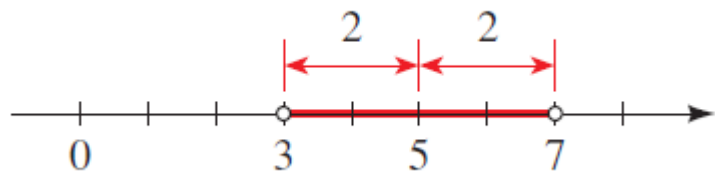
Solve the inequality $|x - 5| < 2$.

SOLUTION 1 The inequality $|x - 5| < 2$ is equivalent to

$$-2 < x - 5 < 2 \quad \text{Property 1}$$

$$3 < x < 7 \quad \text{Add 5}$$

The solution set is the open interval $(3, 7)$.



SOLUTION 2 Geometrically, the solution set consists of all numbers x whose distance from 5 is less than 2. From Figure 9 we see that this is the interval $(3, 7)$.

Solving an absolute value inequality

Solve the inequality $|3x + 2| \geq 4$.

SOLUTION By Property 4 the inequality $|3x + 2| \geq 4$ is equivalent to

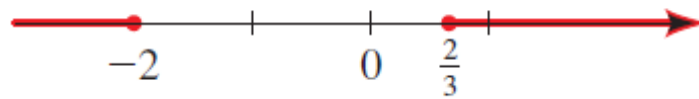
$$3x + 2 \geq 4 \quad \text{or} \quad 3x + 2 \leq -4$$

$$3x \geq 2 \qquad \qquad \qquad 3x \leq -6 \quad \text{Subtract 2}$$

$$x \geq \frac{2}{3} \qquad \qquad \qquad x \leq -2 \quad \text{Divide by 3}$$

So the solution set is

$$\{x \mid x \leq -2 \quad \text{or} \quad x \geq \frac{2}{3}\} = (-\infty, -2] \cup [\frac{2}{3}, \infty)$$



Absolute value inequalities

Solve the absolute value inequality. Express the answer using interval notation and graph the solution set.

(a) $|16x| \leq 8$

(b) $\frac{1}{2}|x| \geq 1$

(c) $|8x + 3| > 12$

(d) $\left| \frac{x + 1}{2} \right| \geq 4$

(e) $3 - |2x + 4| \leq 1$

ACTIVITY 1.3

1.3 Absolute value inequalities

SOLUTIONS

1.3 Absolute value inequalities

1.3 Absolute value inequalities

SOLUTIONS

1.3 Absolute value inequalities

1.3 Absolute value inequalities

SOLUTIONS

2. Coordinate plane, Graphs of lines and circles

Always label your axes



Overview

Coordinate
plane

Graphs of
equations

How to graph regions

Distance
formula

Circles

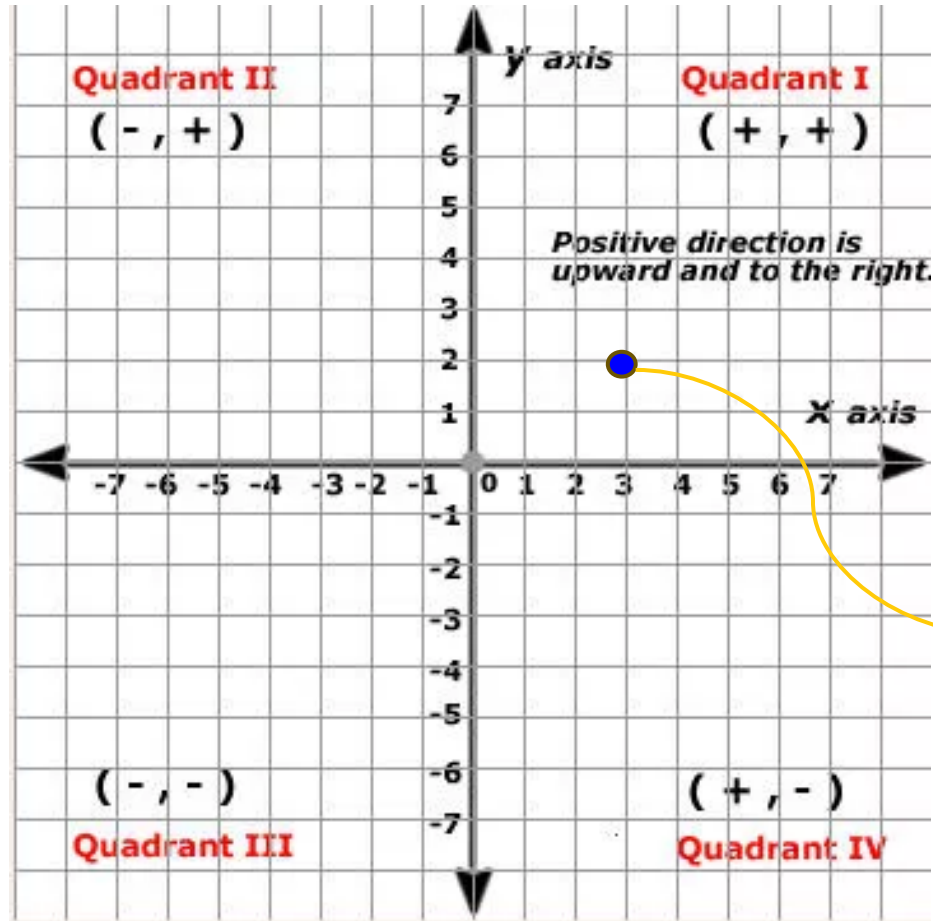
Equation of a circle,
Graphing a circle

Lines

Determining intercepts
Slope of line
Equation of line

The Coordinate plane

Points in a plane can be identified with ordered pairs of numbers to form the coordinate plane or Cartesian plane



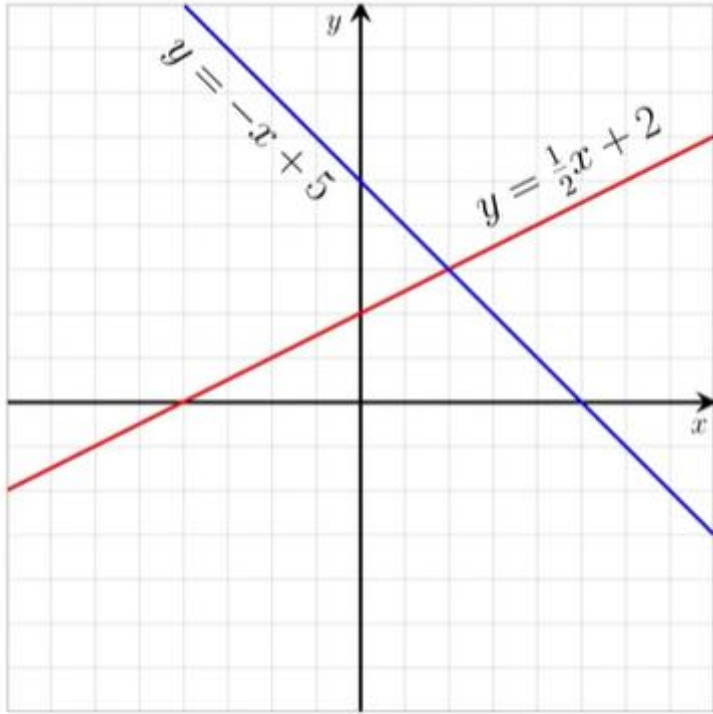
ordered pair of
numbers (a, b)

(3, 2)

(x-coordinate, y-coordinate)

We can think of the coordinates of a point as its "address," because they specify its location in the plane

Graphs

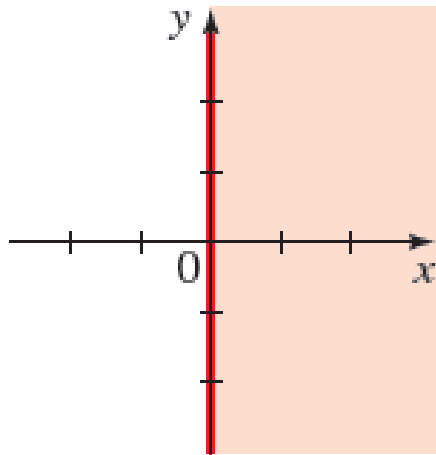


graphs are **visual representations of equations** that help us comprehend the relationship between two variables.

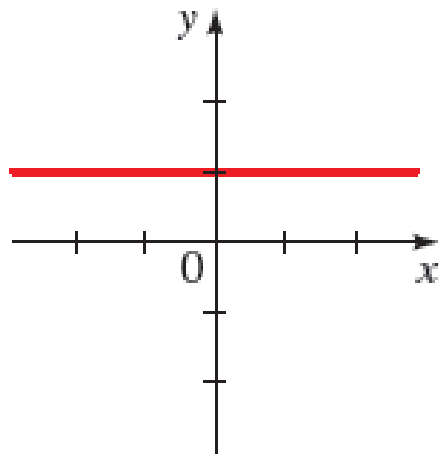
Graphing regions in the coordinate plane

Describe and sketch the regions given by each set.

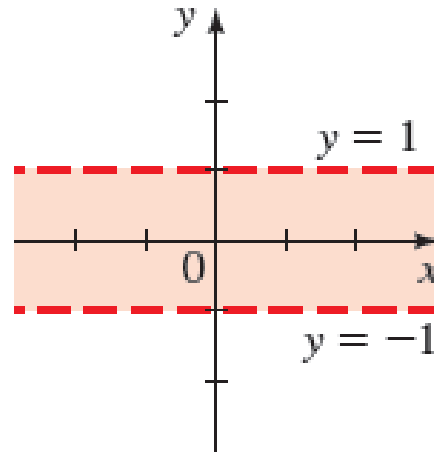
- (a) $\{(x, y) \mid x \geq 0\}$ (b) $\{(x, y) \mid y = 1\}$ (c) $\{(x, y) \mid |y| < 1\}$



(a) $x \geq 0$



(b) $y = 1$



(c) $|y| < 1$

$$|y| < 1 \quad \text{if and only if} \quad -1 < y < 1$$

So the given region consists of those points in the plane whose y -coordinates lie between -1 and 1 . Thus the region consists of all points that lie between (but not on) the horizontal lines $y = 1$ and $y = -1$

Sketching regions

33

Sketch the region given by the set.

(a) $\{(x, y) \mid -3 < x < 3\}$

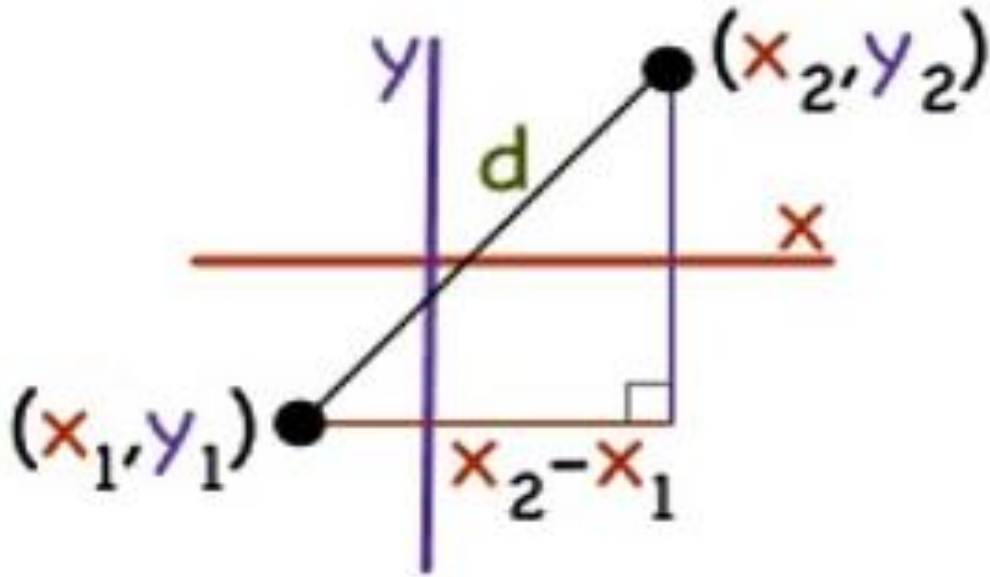
(b) $\{(x, y) \mid |x| \leq 2\}$

(c) $\{(x, y) \mid xy > 0\}$

ACTIVITY 2.1

2.1: Sketching regions

Distance formula



$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

We arrive at this
using the
**Pythagoras
theorem**

Applying the distance formula

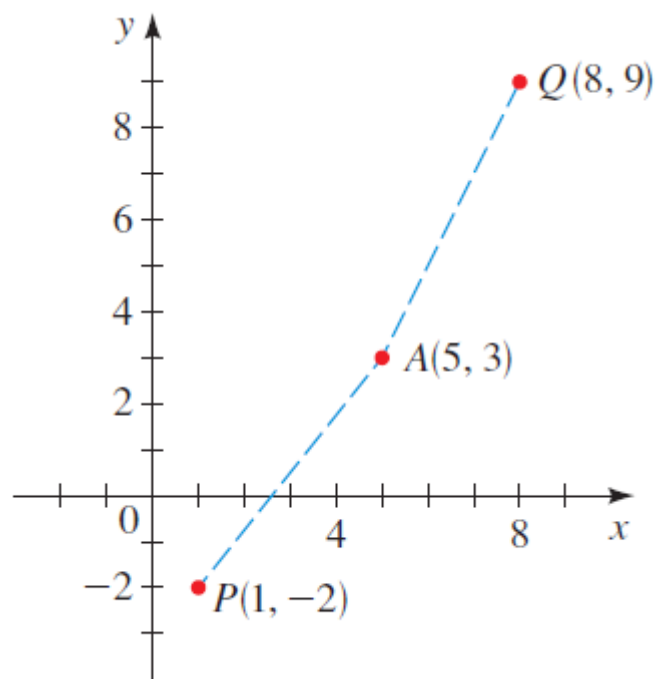
Which of the points $P(1, -2)$ or $Q(8, 9)$ is closer to the point $A(5, 3)$?

SOLUTION By the Distance Formula we have

$$d(P, A) = \sqrt{(5 - 1)^2 + [3 - (-2)]^2} = \sqrt{4^2 + 5^2} = \sqrt{41}$$

$$d(Q, A) = \sqrt{(5 - 8)^2 + (3 - 9)^2} = \sqrt{(-3)^2 + (-6)^2} = \sqrt{45}$$

This shows that $d(P, A) < d(Q, A)$, so P is closer to A



EXAMPLE

Applying the Distance formula

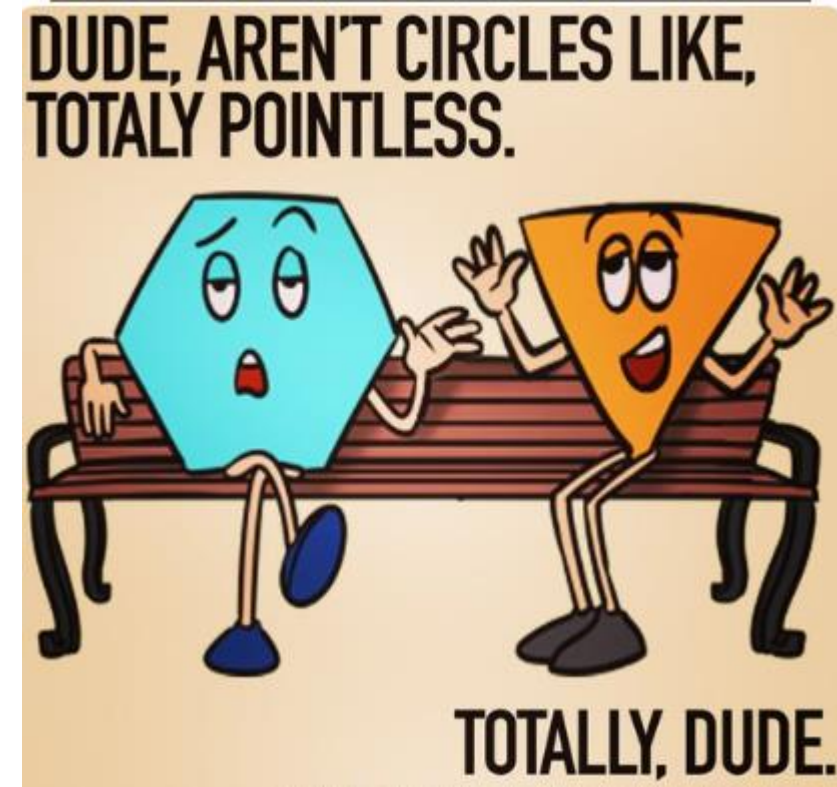
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Show that the points $(7, 3)$ and $(3, 7)$ are the same distance from the origin.

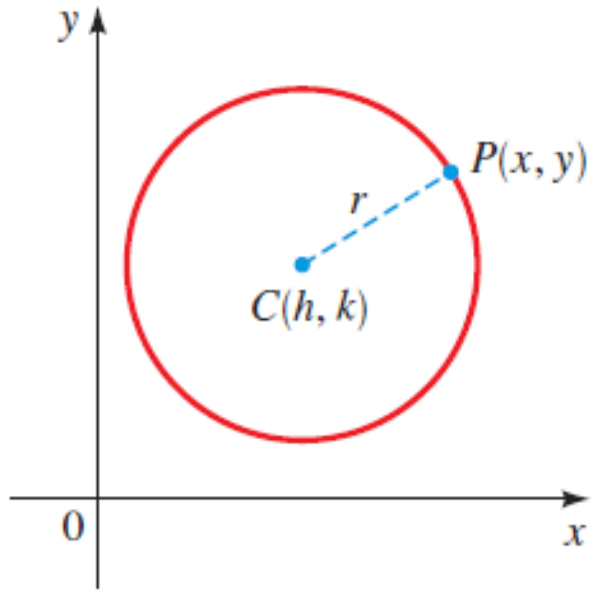
ACTIVITY 2.2

2.2: Applying the Distance formula

Circles



Equation of a circle



By definition the circle is the set of all points $P(x, y)$ whose distance from the center $C(h, k)$ is r .

We use the distance formula to arrive at the equation of a circle

$$\sqrt{(x - h)^2 + (y - k)^2} = r$$

$$(x - h)^2 + (y - k)^2 = r^2 \quad \text{Square each side}$$

EQUATION OF A CIRCLE

An equation of the circle with center (h, k) and radius r is

$$(x - h)^2 + (y - k)^2 = r^2$$

This is called the **standard form** for the equation of the circle. If the center of the circle is the origin $(0, 0)$, then the equation is

$$x^2 + y^2 = r^2$$

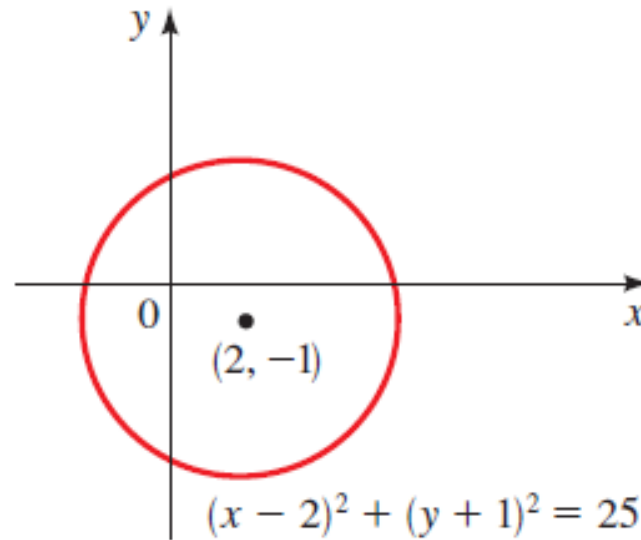
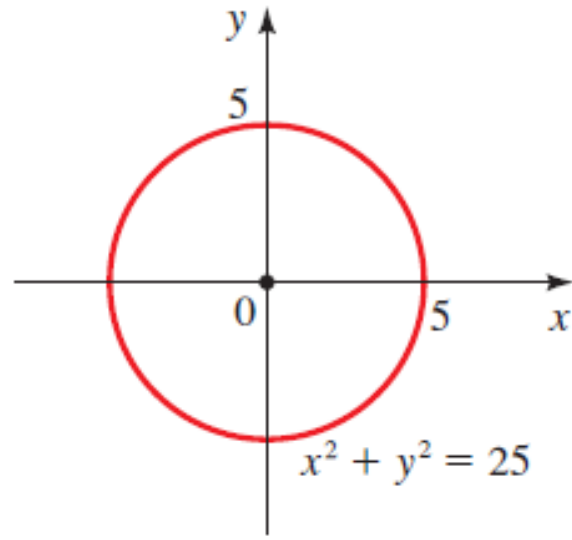
Graphing a circle

Graph each equation.

(a) $x^2 + y^2 = 25$ (b) $(x - 2)^2 + (y + 1)^2 = 25$

SOLUTION

- (a) Rewriting the equation as $x^2 + y^2 = 5^2$, we see that this is an equation of the circle of radius 5 centered at the origin. Its graph is shown in Figure 14.
- (b) Rewriting the equation as $(x - 2)^2 + (y + 1)^2 = 5^2$, we see that this is an equation of the circle of radius 5 centered at $(2, -1)$. Its graph is shown in Figure 15.



Find the centre and radius of circle

Find the center and radius of the circle, and sketch its graph.

(a) $x^2 + y^2 = 5$

(b) $(x + 1)^2 + y^2 = 9$

(c) $(x + 1)^2 + (y + 2)^2 = 36$

(d) $x^2 + y^2 + 6y + 2 = 0$

ACTIVITY 2.3

2.3a Find the centre and radius of circle

SOLUTION

2.3b Find the centre and radius of circle

SOLUTION

2.3c Find the centre and radius of circle

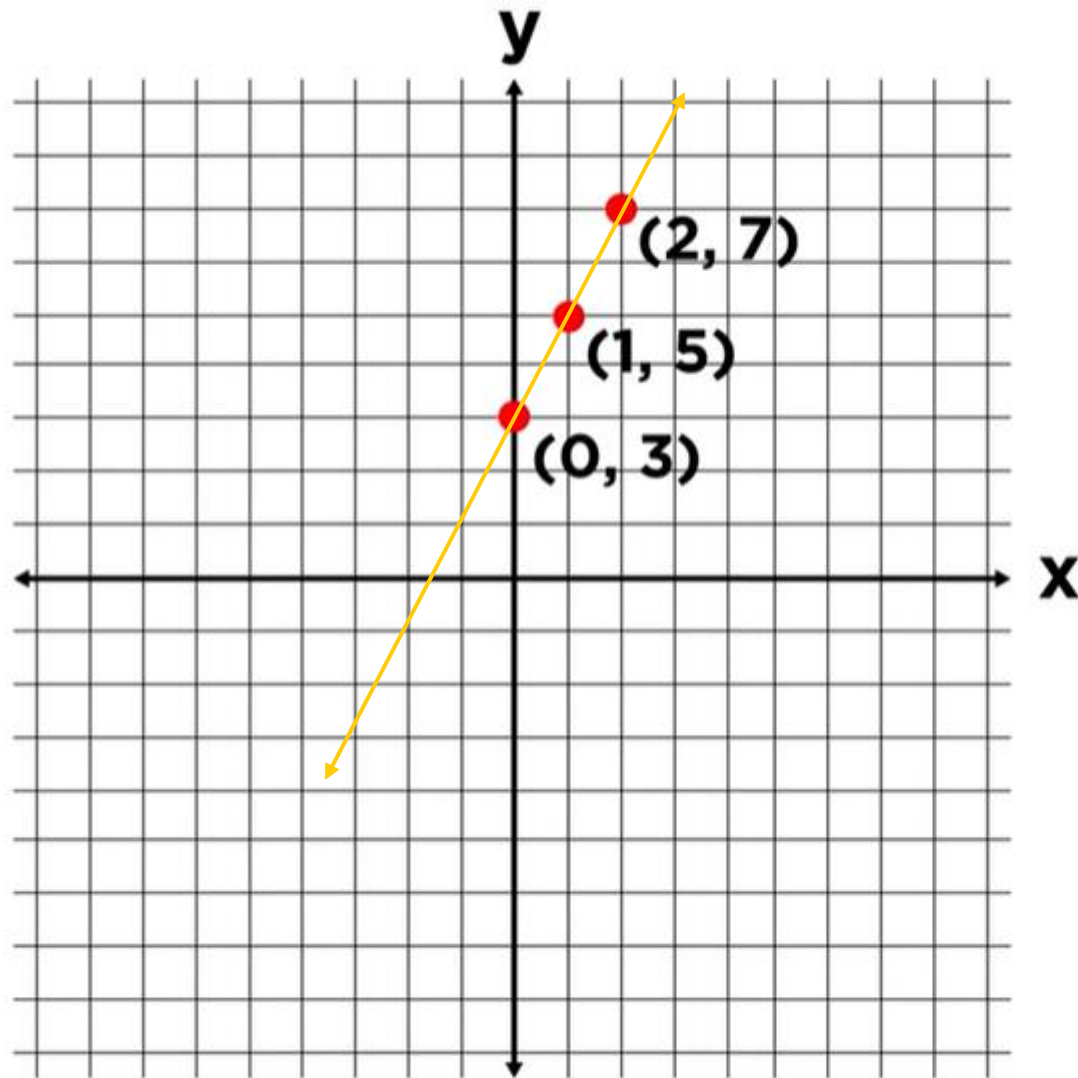
SOLUTION

2.3d Find the centre and radius of circle

SOLUTION

Lines

Graphing equations



$$y = 2x + 3$$

x	y
0	3
1	5
2	7
3	9

each point has **coordinates**
in the form of an **ordered pair**

Definition of intercepts

DEFINITION OF INTERCEPTS

Intercepts

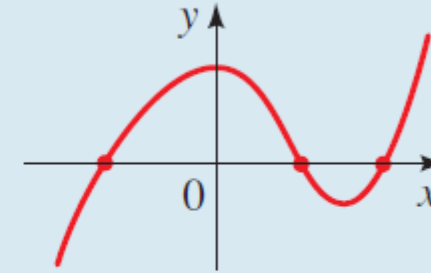
x -intercepts:

The x -coordinates of points where the graph of an equation intersects the x -axis

How to find them

Set $y = 0$ and solve for x

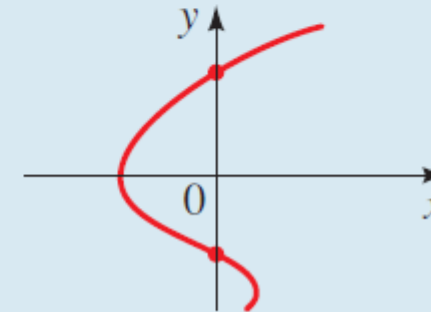
Where they are on the graph



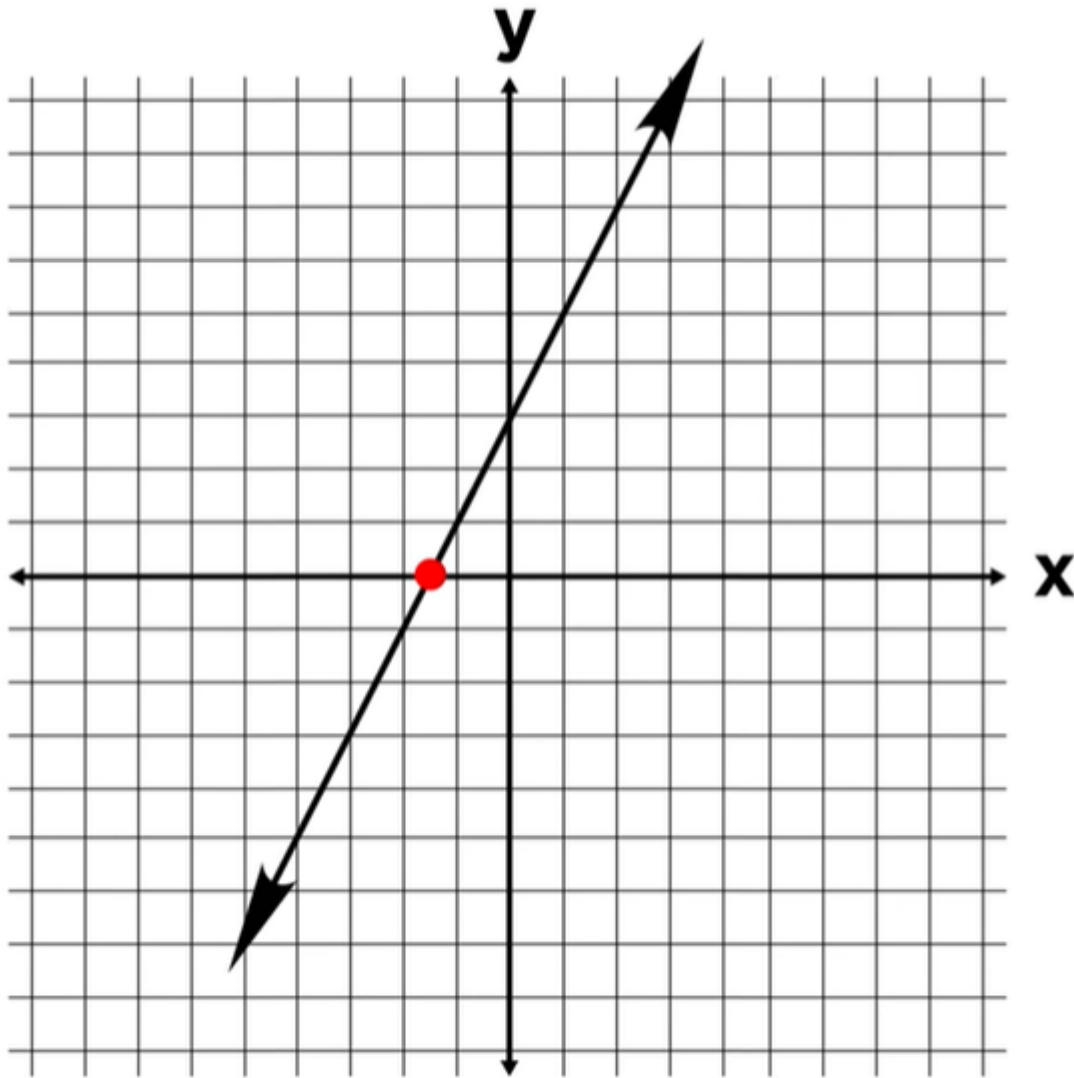
y -intercepts:

The y -coordinates of points where the graph of an equation intersects the y -axis

Set $x = 0$ and solve for y



Determining x intercept



$$y = 2x + 3$$

What is the **x-intercept**?
(value of x when $y = 0$)

$$x = -\frac{3}{2}$$

$$-3/2 = x$$

Find the x- and y-intercepts of the graph of the equation

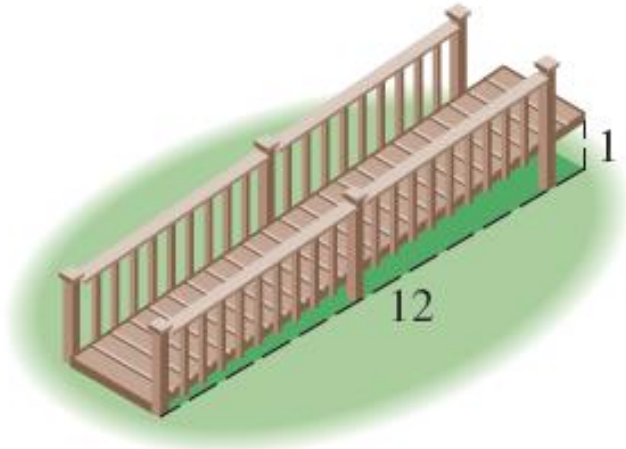
(a) $4x^2 + 25y^2 = 100$

(b) $x^2 - xy + 3y = 1$

2.4a: Intercepts

2.4b: Intercepts

How to measure the 'steepness' of a line?



Slope of a ramp
Slope = $\frac{1}{12}$



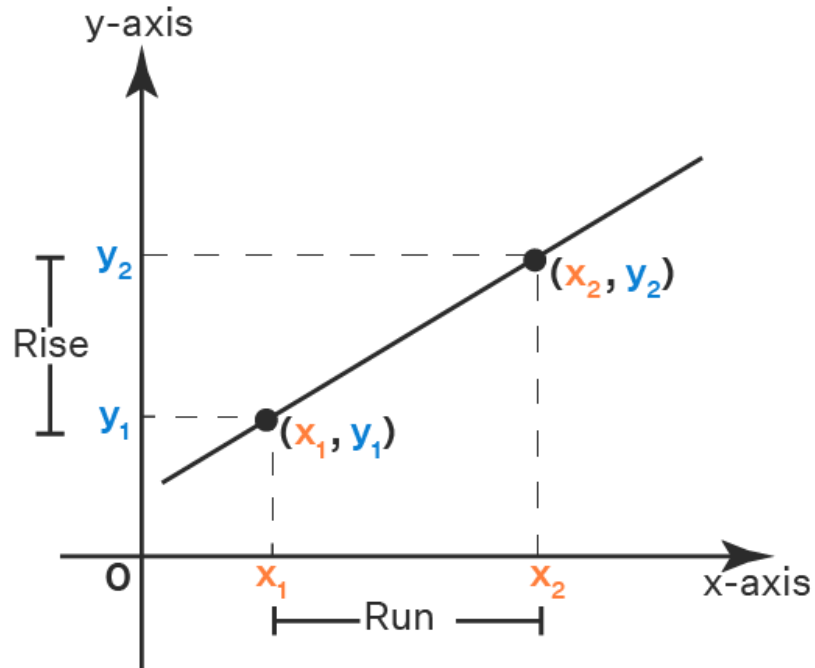
Pitch of a roof
Slope = $\frac{1}{3}$



Grade of a road
Slope = $\frac{8}{100}$

The concept of '**slope**' helps us in measuring the steepness of lines

Lines and slopes of lines



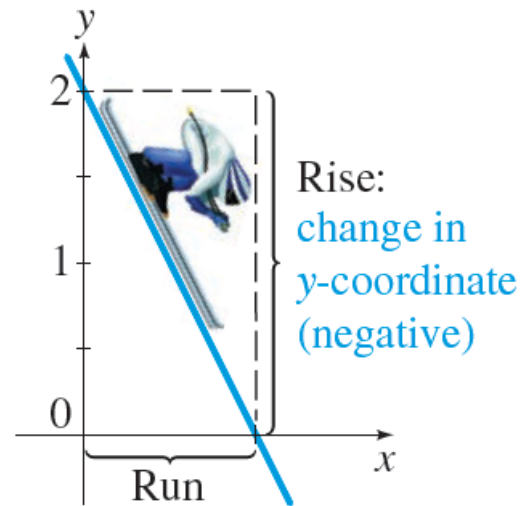
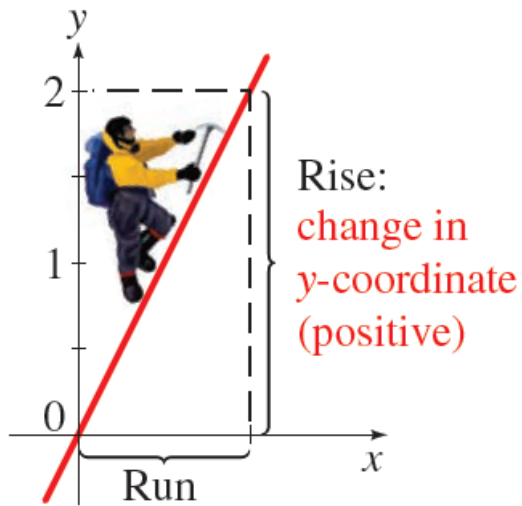
run is the change in the x-coordinate

rise is the corresponding change in the y-coordinate

$$\text{slope} = \frac{\text{rise}}{\text{run}}$$

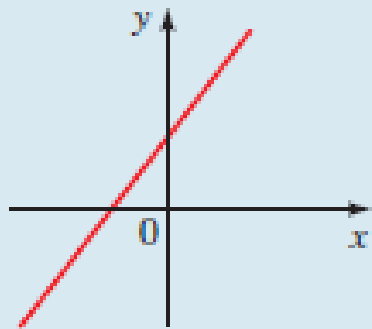
Slope measures how **quickly** a line rises (or falls) as we move **from left to right**.

Positive and negative slope

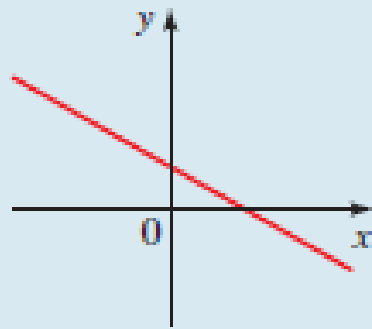


SLOPE OF A LINE

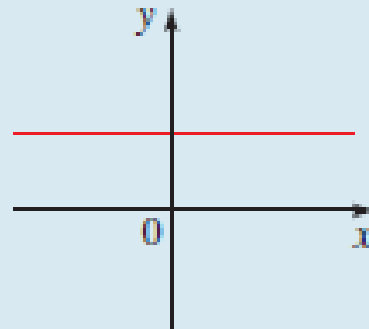
Positive Slope



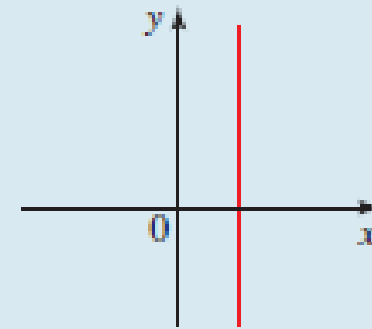
Negative Slope



Zero Slope



No Slope



Slope: Formal definition

SLOPE OF A LINE

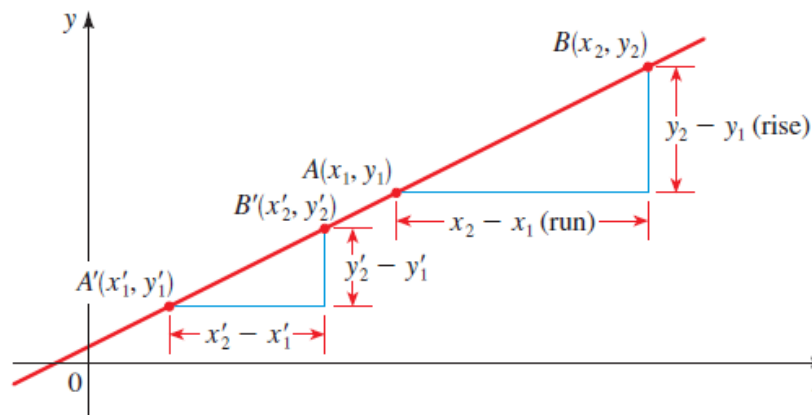
The **slope** m of a nonvertical line that passes through the points $A(x_1, y_1)$ and $B(x_2, y_2)$ is

$$m = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1}$$

The slope of a vertical line is not defined.

The slope is **independent** of which two points are chosen on the line.

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{y'_2 - y'_1}{x'_2 - x'_1}$$



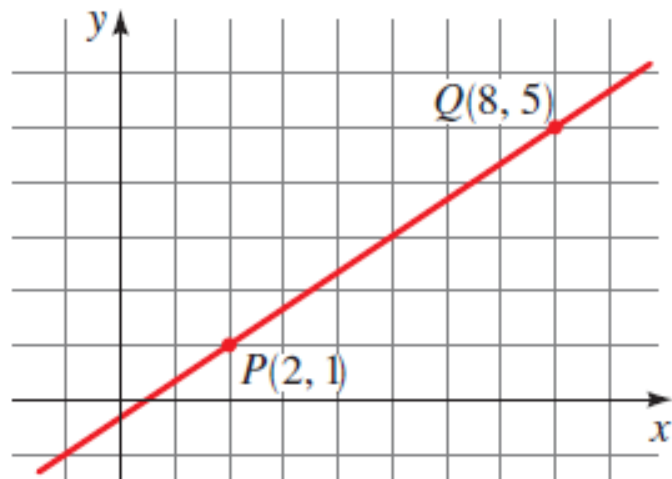
Finding the slope of a line thru two points

Find the slope of the line that passes through the points $P(2, 1)$ and $Q(8, 5)$.

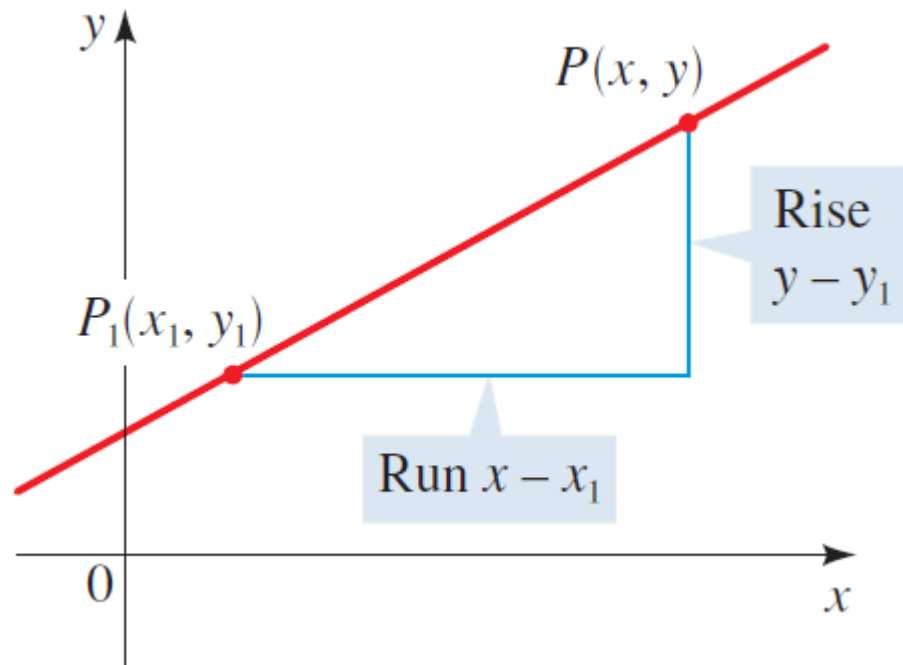
SOLUTION Since any two different points determine a line, only one line passes through these two points. From the definition the slope is

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{5 - 1}{8 - 2} = \frac{4}{6} = \frac{2}{3}$$

This says that for every 3 units we move to the right, the line rises 2 units.



Point-Slope Form of the Equation of a line



$$\frac{y - y_1}{x - x_1} = m$$

POINT-SLOPE FORM OF THE EQUATION OF A LINE

An equation of the line that passes through the point (x_1, y_1) and has slope m is

$$y - y_1 = m(x - x_1)$$

Finding an Equation of a Line with Given Point and Slope

- (a) Find an equation of the line through $(1, -3)$ with slope $-\frac{1}{2}$.
 (b) Sketch the line.

SOLUTION

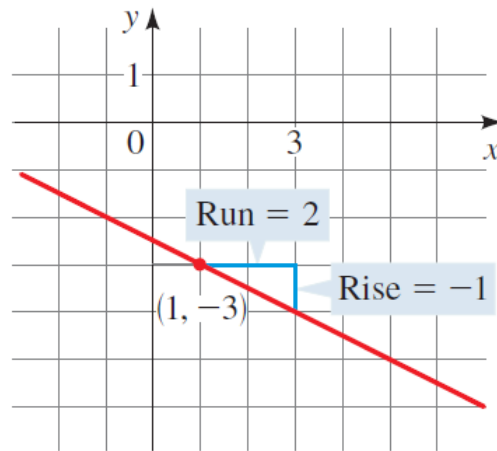
- (a) Using the point-slope form with $m = -\frac{1}{2}$, $x_1 = 1$, and $y_1 = -3$, we obtain an equation of the line as

$$y + 3 = -\frac{1}{2}(x - 1) \quad \text{Slope } m = -\frac{1}{2}, \text{ point } (1, -3)$$

$$2y + 6 = -x + 1 \quad \text{Multiply by 2}$$

$$x + 2y + 5 = 0 \quad \text{Rearrange}$$

- (b) The fact that the slope is $-\frac{1}{2}$ tells us that when we move to the right 2 units, the line drops 1 unit. This enables us to sketch the line



Finding an Equation of a Line Through Two Given Points

Find an equation of the line through the points $(-1, 2)$ and $(3, -4)$.

SOLUTION The slope of the line is

$$m = \frac{-4 - 2}{3 - (-1)} = -\frac{6}{4} = -\frac{3}{2}$$

Using the point-slope form with $x_1 = -1$ and $y_1 = 2$, we obtain

$$y - 2 = -\frac{3}{2}(x + 1) \quad \text{Slope } m = -\frac{3}{2}, \text{ point } (-1, 2)$$

$$2y - 4 = -3x - 3 \quad \text{Multiply by 2}$$

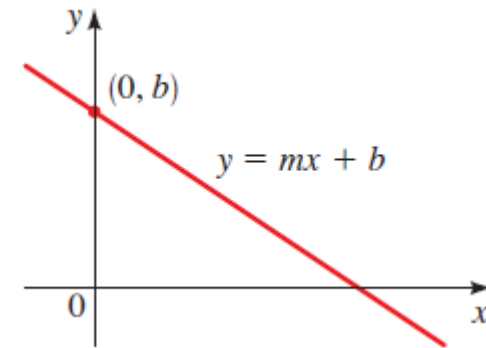
$$3x + 2y - 1 = 0 \quad \text{Rearrange}$$

Slope-Intercept Form of the Equation of a Line

Suppose a nonvertical line has slope m and y-intercept b (see Figure 7). This means that the line intersects the y-axis at the point $(0, b)$, so the point-slope form of the equation of the line, with $x = 0$ and $y = b$, becomes

$$y - b = m(x - 0)$$

This simplifies to $y = mx + b$, which is called the **slope-intercept form** of the equation of a line.



SLOPE-INTERCEPT FORM OF THE EQUATION OF A LINE

An equation of the line that has slope m and y-intercept b is

$$y = mx + b$$

EXAMPLE

Lines in slope intercept form

- (a) Find an equation of the line with slope 3 and y-intercept -2 .
- (b) Find the slope and y-intercept of the line $3y - 2x = 1$.

SOLUTION

- (a) Since $m = 3$ and $b = -2$, from the slope-intercept form of the equation of a line we get

$$y = 3x - 2$$

- (b) We first write the equation in the form $y = mx + b$.

$$3y - 2x = 1$$

$$3y = 2x + 1 \quad \text{Add } 2x$$

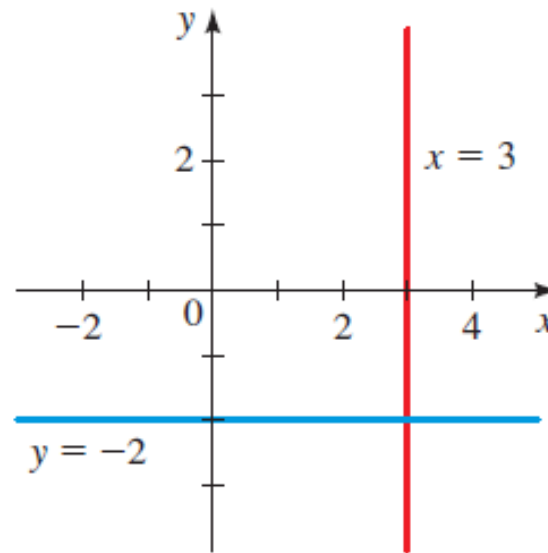
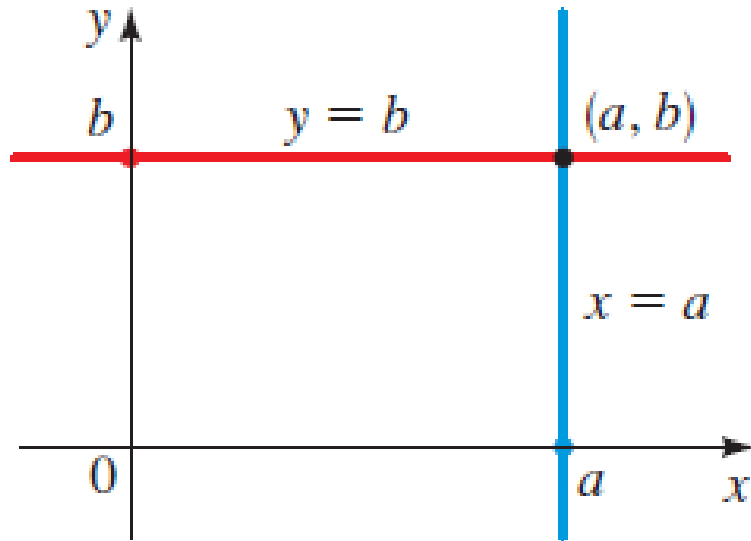
$$y = \frac{2}{3}x + \frac{1}{3} \quad \text{Divide by } 3$$

From the slope-intercept form of the equation of a line, we see that the slope is $m = \frac{2}{3}$ and the y-intercept is $b = \frac{1}{3}$.

Vertical and horizontal lines

VERTICAL AND HORIZONTAL LINES

- An equation of the vertical line through (a, b) is $x = a$.
- An equation of the horizontal line through (a, b) is $y = b$.



Find equation of line

Find an equation of the line that satisfies the given conditions.

(a) Slope $\frac{2}{5}$; y -intercept 4

(b) Through $(-1, -2)$ and $(4, 3)$

ACTIVITY 2.5

2.5a Find equation of line

SOLUTION

2.5b Find equation of line

SOLUTION

Using Slopes and y-Intercepts to Graph Lines

Find the slope and y-intercept of the line, and draw its graph.

(a) $y = \frac{2}{3}x - 2$

(b) $2x - 5y = 0$

ACTIVITY 2.6

2.6a Using Slopes and y-Intercepts to Graph Lines

SOLUTIONS

2.6b Using Slopes and y-Intercepts to Graph Lines

SOLUTION